



Value at Risk of Bonds

Risk Management in Financial Institutions

Lecture 3

Béla ÖCSI

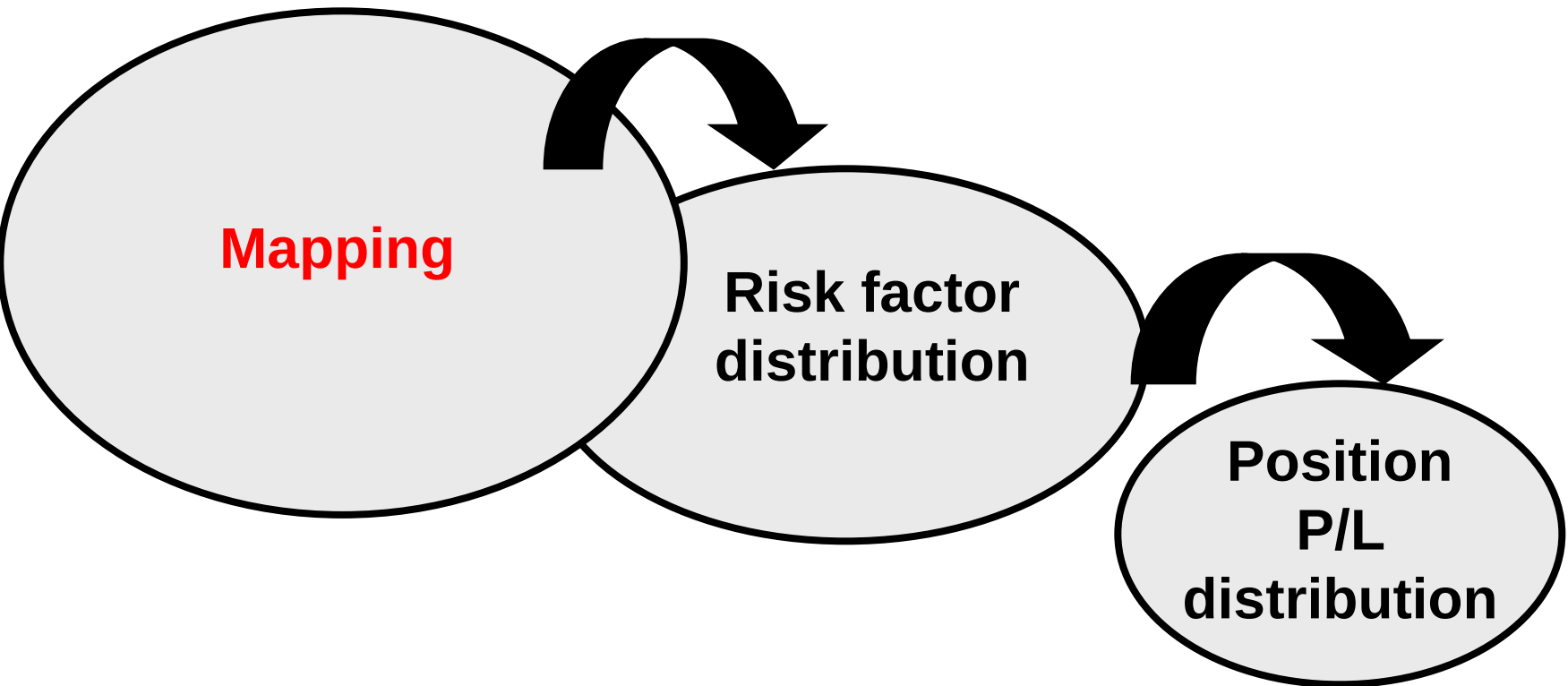
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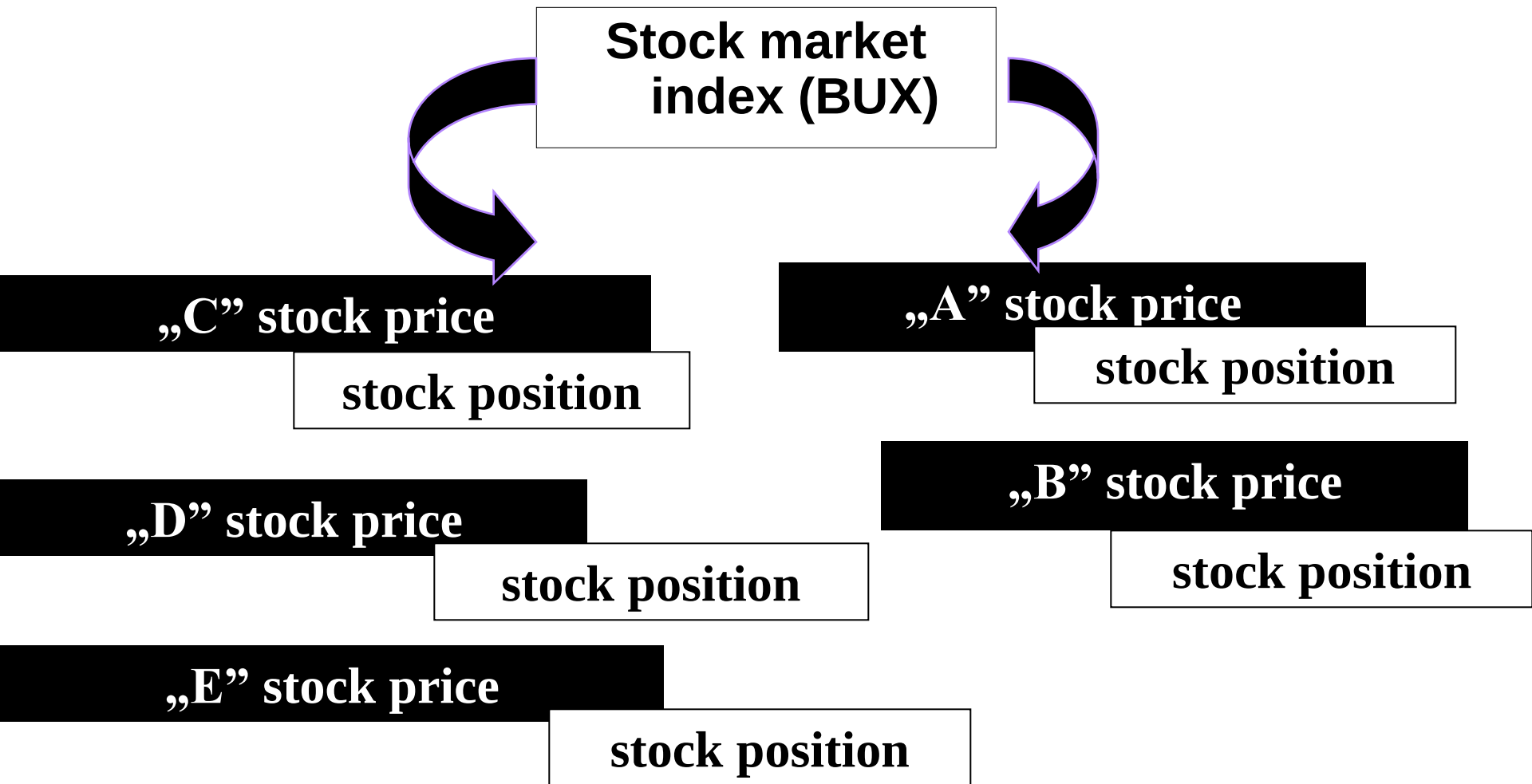


Logical model for determining the Value at Risk



Mapping the position: the key to VAR is tracing back to the risk factor

1. „Stock market index mapping”



In practice, stock market indices are not mapped, but used as risk factors (and some stocks are mapped to indices using factors models)

Composition of the BUX index

Indexkosarak - Bet site

bet.hu/oldalak/indexkosarak

KERESÉS

UK

KERESKEDÉSI ADATOKBEFEKTETŐKKIBOCSÁTÓKTŐZSDETAGOKVENDOROKRÓLUNKBÉT ESG

BUX KOSÁR ÖSSZETÉTELE

BUMIX KOSÁR ÖSSZETÉTELE

CETOP KOSÁR ÖSSZETÉTELE

Dátum: 2021.09.30. AF érték: 0,1984778507 Bázisérték: 1 000 Báziskapitalizáció: 14 639 314 708 Záróérték: 52 854,65 Változás: 850,05 Változás (%): 1,63

Részvény	ISIN kód	Tőzsdére bev. menny.	Közkészhiányad	Súlyozási tényező	Záróár (Ft)	Számláló (Ft)	Súly a kosárban (%)
4IG	HU0000167788	99 207 921	0,3559	1,000000	990	34 955 018 093,06	0,90
AKKO INVEST	HU0000170824	33 355 200	0,5890	1,000000	418	8 212 116 950,40	0,21
ALTEO	HU0000155726	19 386 274	0,3668	1,000000	1 360	9 670 804 012,35	0,25
ANY	HU0000093257	14 794 650	0,7806	1,000000	1 580	18 246 951 988,20	0,47
APPENINN	HU0000102132	47 371 419	0,5258	1,000000	283	7 048 933 467,19	0,18
AUTOWALLIS	HU0000164504	324 313 680	0,3036	1,000000	127	12 504 627 422,50	0,32
CIGPANNONIA	HU0000180112	94 428 260	0,4460	1,000000	325	13 687 376 287,00	0,35
GRAPHISOFT PARK SE	HU0000083696	10 631 674	0,6427	1,000000	3 750	25 623 663 299,25	0,66
MAGYAR TELEKOM	HU0000073507	1 042 742 543	0,4079	1,000000	423,5	180 129 238 373,19	4,62
MASTERPLAST	HU0000093943	14 601 279	0,3683	1,010290	4 400	23 905 143 174,28	0,61
MOL	HU0000153937	819 424 824	0,6237	0,762292	2 584	1 006 696 901 510,95	25,82
OPUS	HU0000110226	701 646 050	0,3749	1,000000	253,5	66 682 440 900,76	1,71
OTP BANK	HU0000061726	280 000 010	0,7407	0,425843	18 200	1 607 390 111 276,39	41,23
PANNERGY	HU0000089867	21 054 655	0,5001	1,000000	898	9 455 430 803,02	0,24
RICHTER GEDEON	HU0000123096	186 374 860	0,6968	0,782725	8 485	862 494 877 010,45	22,12
WABERERS	HU0000120720	17 693 734	0,2801	1,000000	2 370	11 745 755 297,36	0,30

kon 8:30-kor folytatódik. | A kereskedés tőzsdenapokon 8:30-kor folytatódik. | A kereskedés tőzsdenapokon 8:30-kor folytatódik. | A kereskedés tőzsdenapokon 8:30-kor folytatódik. | A kereskedés tőzsdenapokon 8:30-kor folytatódik.

14:222020. 09. 25.

▼ MOL 2 970 -0,67% | ▲ MTELEKOM 400 +0,25% | ■ OPUS 541 0,00% | ▼ OTP 9 845 -0,10% | ▲ PANNERGY 690 +1,47% | ■ RABA 1 300 0,00% | ■ RICHTER 5 190 0,00% | ■ TAI

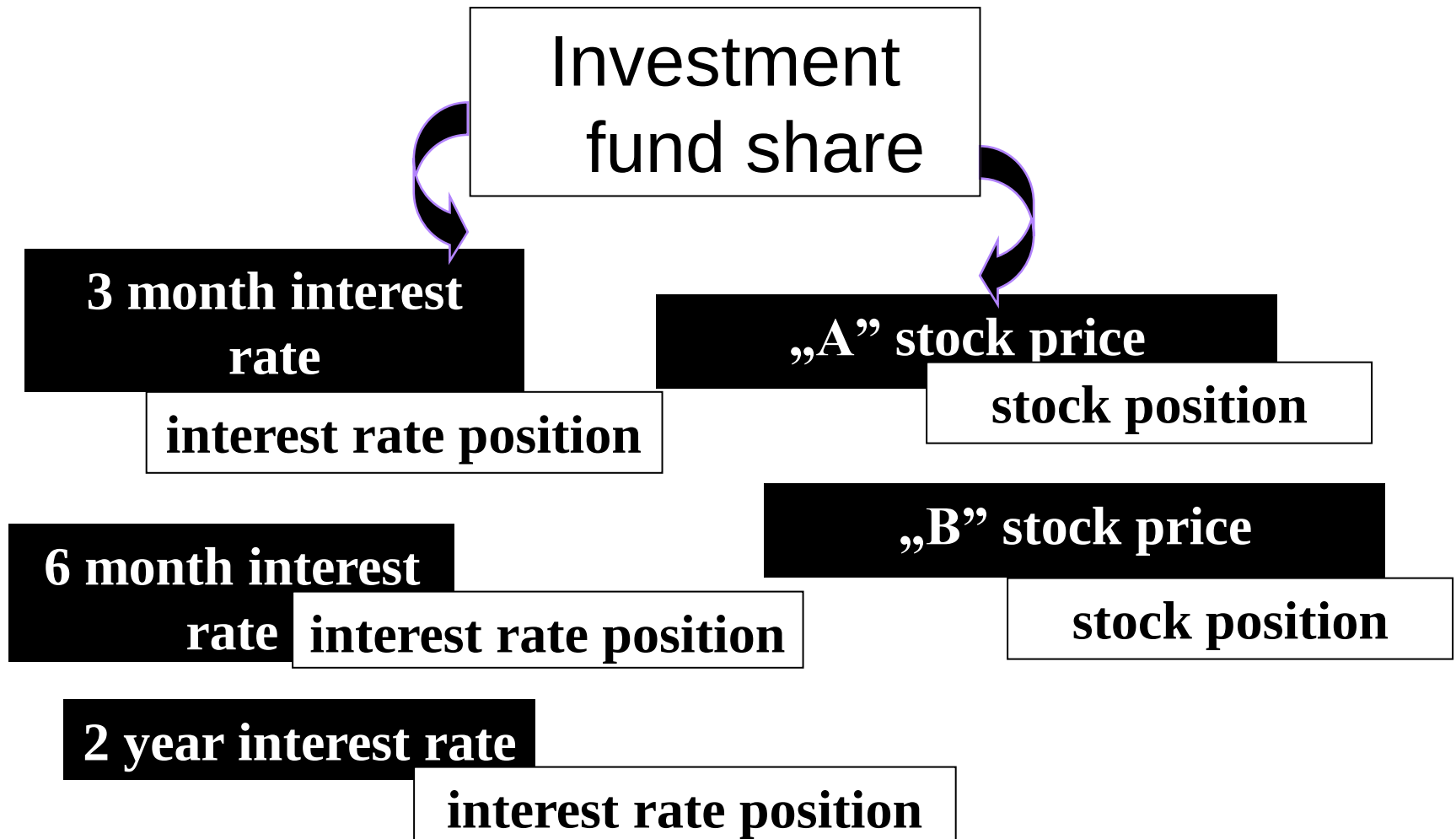
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Composition of the DJIA index

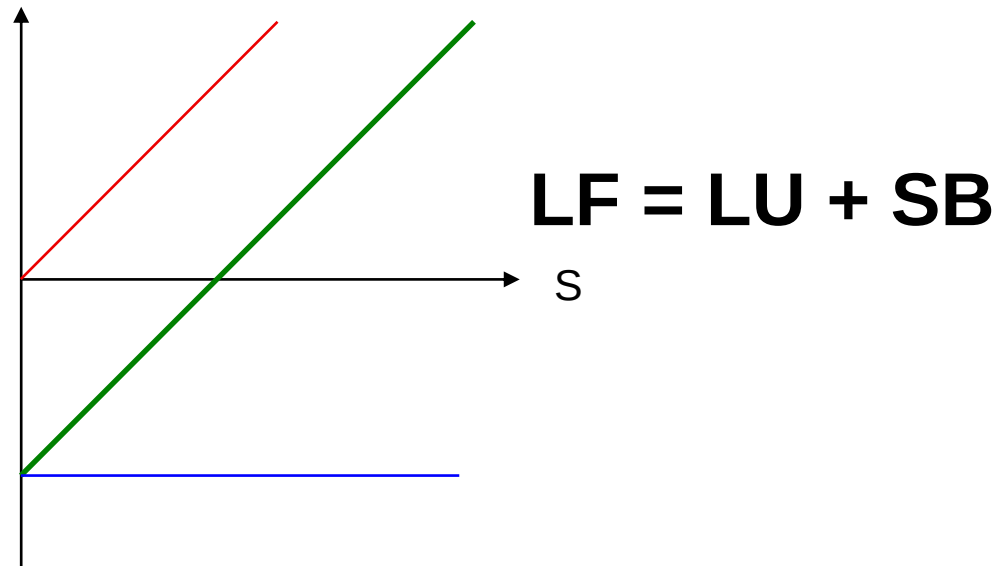
Company	Exchange	Symbol	Industry	Date added	Notes	Index weighting
UnitedHealth Group	NYSE	UNH	Managed health care	2012-09-24		7.88%
Goldman Sachs	NYSE	GS	Financial services	2013-09-20		7.60%
The Home Depot	NYSE	HD	Retailing	1999-11-01		6.05%
Microsoft	NASDAQ	MSFT	Information technology	1999-11-01		5.63%
Salesforce	NYSE	CRM	Information technology	2020-08-31		4.83%
McDonald's	NYSE	MCD	Food industry	1985-10-30		4.44%
Visa Inc.	NYSE	V	Financial services	2013-09-20		4.36%
Honeywell	NASDAQ	HON	Conglomerate	2020-08-31		4.29%
Amgen	NASDAQ	AMGN	Pharmaceutical industry	2020-08-31		4.18%
Boeing	NYSE	BA	Aerospace and defense	1987-03-12		4.12%
Caterpillar Inc.	NYSE	CAT	Construction and Mining	1991-05-06		3.96%
3M	NYSE	MMM	Conglomerate	1976-08-09	As Minnesota Mining and Manufacturing	3.62%
The Walt Disney Company	NYSE	DIS	Broadcasting and entertainment	1991-05-06		3.32%
Johnson & Johnson	NYSE	JNJ	Pharmaceutical industry	1997-03-17		3.26%
Nike, Inc.	NYSE	NKE	Apparel	2013-09-20		3.16%
American Express	NYSE	AXP	Financial services	1982-08-30		3.00%
The Travelers Companies	NYSE	TRV	Financial services	2009-06-08		2.99%
JPMorgan Chase	NYSE	JPM	Financial services	1991-05-06		2.93%
Apple Inc.	NASDAQ	AAPL	Information technology	2015-03-19		2.78%
Walmart	NYSE	WMT	Retailing	1997-03-17		2.77%
Procter & Gamble	NYSE	PG	Fast-moving consumer goods	1932-05-26		2.66%
IBM	NYSE	IBM	Information technology	1979-06-29	Also 1932-05-26 to 1939-03-04	2.60%
Chevron Corporation	NYSE	CVX	Petroleum industry	2008-02-19	Also 1930-07-18 to 1999-11-01	1.82%
Merck & Co.	NYSE	MRK	Pharmaceutical industry	1979-06-29		1.45%
Dow Inc.	NYSE	DOW	Chemical industry	2019-04-02		1.18%
Cisco Systems	NASDAQ	CSCO	Information technology	2009-06-08		1.10%
The Coca-Cola Company	NYSE	KO	Food industry	1987-03-12	Also 1932-05-26 to 1935-11-20	1.04%
Verizon Communications	NYSE	VZ	Telecommunication	2004-04-08		1.02%
Intel	NASDAQ	INTC	Information technology	1999-11-01		1.00%
Walgreens Boots Alliance	NASDAQ	WBA	Retailing	2018-06-26		0.90%

2. Investment fund shares mapping



3. Mapping forward products

- Mapping
 - Underlying product
 - Interest rate position



- Value of forward contract

- Share:

$$f = S - PK$$

, where

- S: exchange rate,
 - P: local currency DF,
 - K: strike price

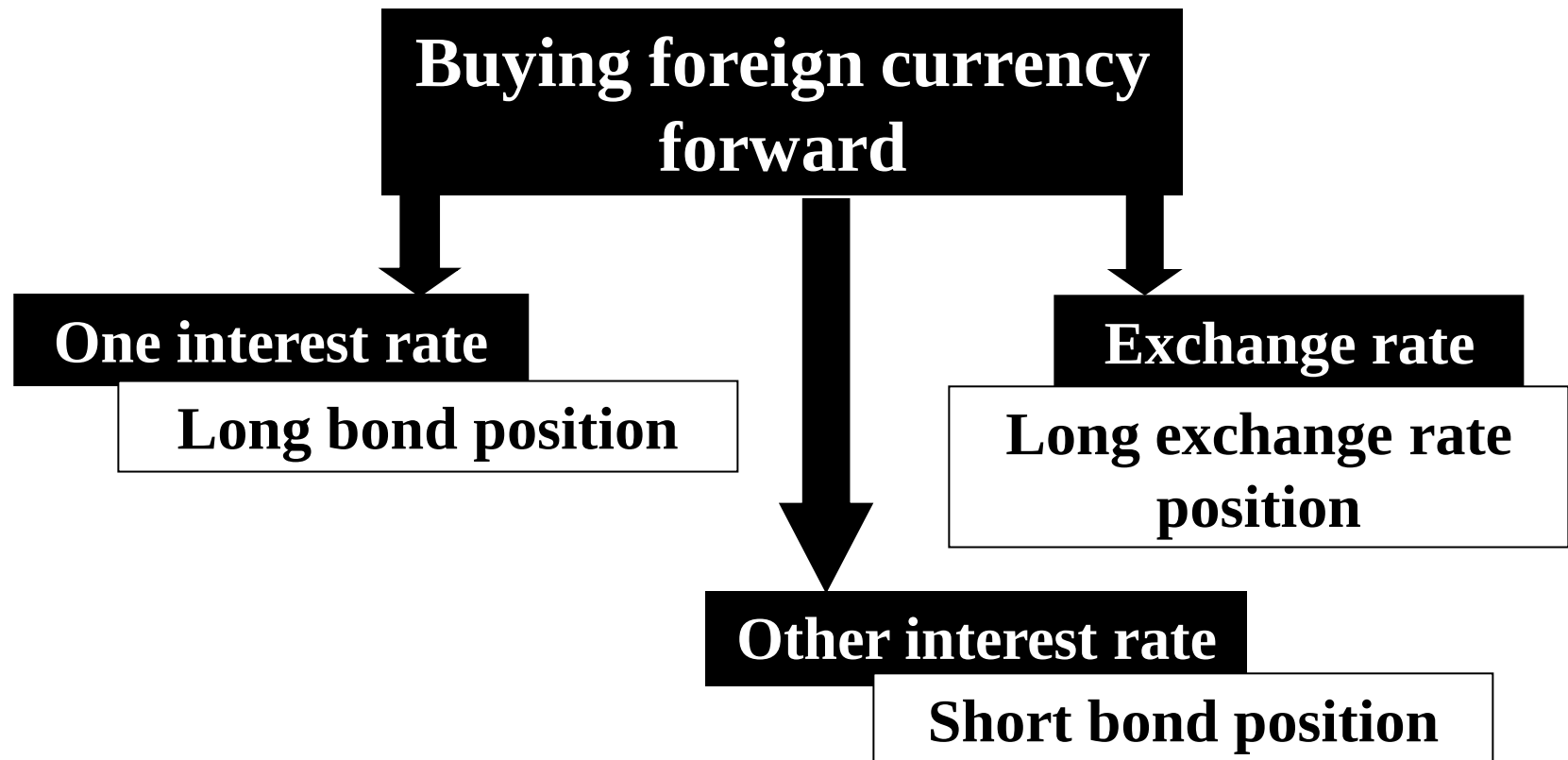
- Foreign currency:

$$f = SQ - PK$$

, ahol

- S: exchange rate,
 - Q: foreign currency DF,
 - P: local currency DF,
 - K: strike rate

3. Mapping of foreign currency forwards



3. Mapping of foreign currency forwards

$$f = SQ - PK$$

Exchange rate of the underlying product

Price of a foreign currency bond with a face value of one unit

Price of forint bond with face value K

○ That is

Long underlying + Long bond + Short bond

4. Mapping bonds to risk factor

Bond price without arbitrage, P

$$= \sum c_t \cdot (1+r_t)^{-t}$$

Risk factors: r_t yields (yield curve points)

Problem: too much r_t (every day for zero-coupon curve)

For how much and to what yield can exchange rate changes be mapped?



1. Mapping to a single interest rate - duration
(„ r ” = YTM)



2. Mapping to an interest rate series ($r_1, r_2, \dots$ = yield curve)

Alternatives for Handling Interest Rates

- Duration approach: Linear relation between ΔP and Δy ($y=r$) but assumes parallel shifts)
- Cash flow mapping: Variables are zero-coupon bond prices with about 10 different maturities
- Principal components analysis: 2 or 3 independent shifts with their own volatilities

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Duration-based mapping: approximation

We neglect convexity - inaccurate for long maturities

$$\Delta V = f(x_0 + \Delta x) - f(x_0) = \underbrace{\frac{df}{dx} dx}_{\text{Duration (average time)}} + \underbrace{\frac{1}{2} \frac{d^2 f}{dx^2} dx^2}_{\text{Convexity (curvature)}} + \circ$$

(Taylor series). Switching to the usual notation ($V=P$, $x=r$):

$$\Delta P = -D_{\text{mod}} \cdot P \cdot \Delta r + \left(\frac{1}{2} \cdot Cx \cdot P \cdot \Delta r^2 + \circ \right), \text{ where}$$

$$D_{\text{mod}} = \frac{D}{(1 + YTM)}$$

P: bond price

Delta-
normal
case!

$$VaR_{\text{bond}} = D_{\text{mod}} * P * VaR(\%)_{\text{yield}}$$

Duration (page 182)

- Duration of a bond that provides cash flow c_i at time t_i is

$$\sum_{i=1}^n t_i \left(\frac{c_i e^{-yt_i}}{B} \right)$$

where B is its price and y is its yield (continuously compounded)

- This leads to

$$\frac{\Delta B}{B} = -D\Delta y$$

Calculation of Duration for a 3-year bond paying a coupon 10%. Bond yield=12%. (Table 9.3, page 183)

Time (yrs)	Cash Flow (\$)	PV (\$)	Weight	Time × Weight
0.5	5	4.709	0.050	0.025
1.0	5	4.435	0.047	0.047
1.5	5	4.176	0.044	0.066
2.0	5	3.933	0.042	0.083
2.5	5	3.704	0.039	0.098
3.0	105	73.256	0.778	2.333
Total	130	94.213	1.000	2.653

Duration Continued

- When the yield y is expressed with compounding m times per year

$$\Delta B = -\frac{BD\Delta y}{1 + y/m}$$

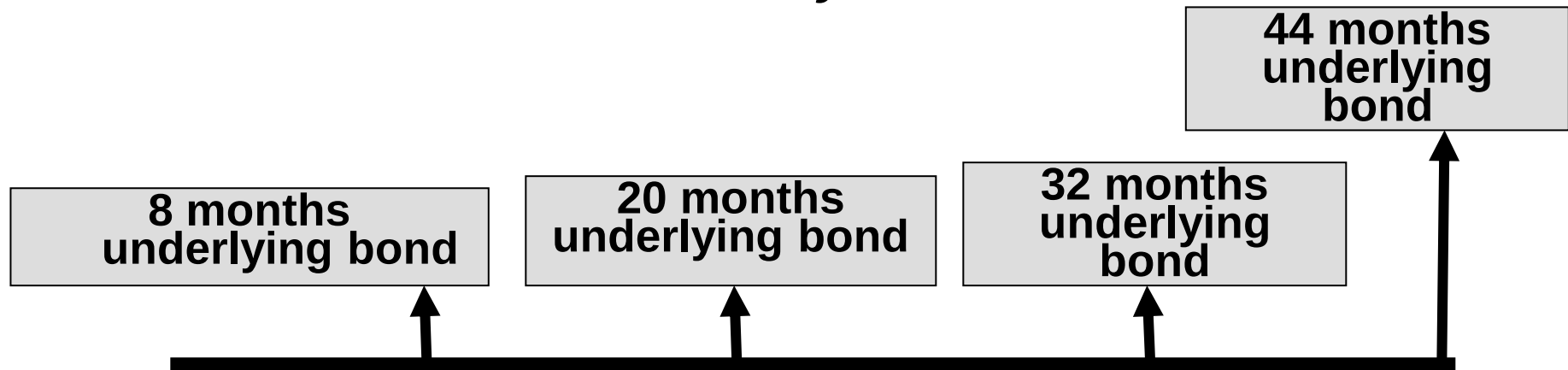
- The expression

$$\frac{D}{1 + y/m}$$

is referred to as the “modified duration”

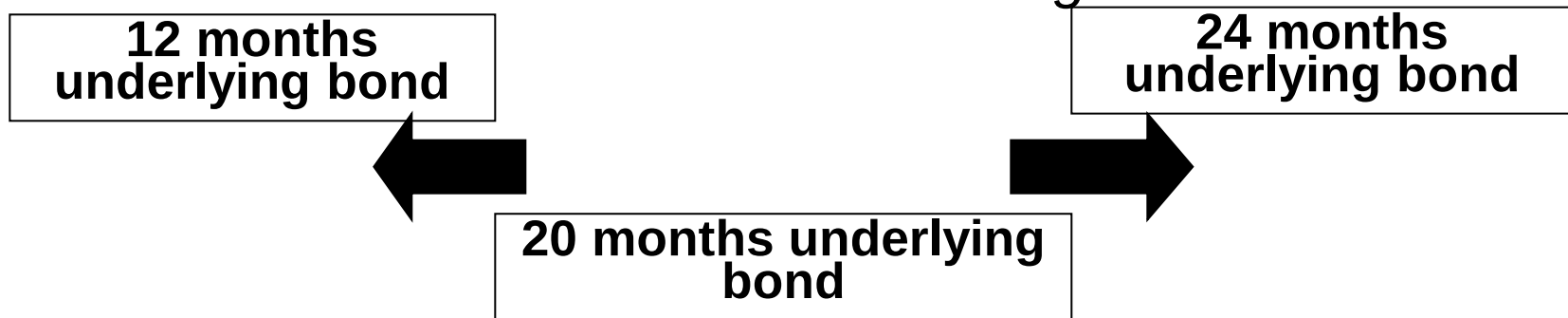
Cash flow based decomposition

Decomposition into underlying bonds with a standard maturity



Decomposition of underlying bonds into standard positions

So that neither the present value (PV) nor the risk (variance) of the new CFs change



Handling Interest Rates: Cash Flow Mapping (page 307-309)

- We choose as market variables zero-coupon bond prices with standard maturities (1mm, 3mm, 6mm, 1yr, 2yr, 5yr, 7yr, 10yr, 30yr)
- Suppose that the 5yr rate is 6% and the 7yr rate is 7% and we will receive a cash flow of \$10,000 in 6.5 years.
- The volatilities per day of the 5yr and 7yr bonds are 0.50% and 0.58% respectively

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Example continued

- We interpolate between the 5yr rate of 6% and the 7yr rate of 7% to get a 6.5yr rate of 6.75%
- The PV of the \$10,000 cash flow is

$$\frac{10,000}{1.0675^{6.5}} = 6,540$$

Example continued

- We interpolate between the 0.5% volatility for the 5yr bond price and the 0.58% volatility for the 7yr bond price to get 0.56% as the volatility for the 6.5yr bond
- We allocate α of the PV to the 5yr bond and $(1 - \alpha)$ of the PV to the 7yr bond

Example continued

- Suppose that the correlation between movement in the 5yr and 7yr bond prices is 0.6

- To match variances

$$0.56^2 = 0.5^2 \alpha^2 + 0.58^2 (1 - \alpha)^2 + 2 \times 0.6 \times 0.5 \times 0.58 \times \alpha (1 - \alpha)$$

- This gives $\alpha=0.074$

Example continued

The value of 6,540 received in 6.5 years

$$6,540 \times 0.074 = \$484$$

in 5 years and by

$$6,540 \times 0.926 = \$6,056$$

in 7 years.

This cash flow mapping preserves value and variance